

Test of Chemistry

Section A

Q1. (4)

Q2. (2)

Q3. (2)

Q4. (1)

Q5. (1)

Q6. (4)

Q7. (2)

Q8. (1)

Q9. (2)

Q10. (4)

Q11. (1)

Q12. (3)

Q13. (3)

Q14. (2)

Q15. (3)

Q16. (a) Y

(b) L

(c) Halogen

(d) Boron and Nitrogen

(e) Oxygen

(ii) (a) ~~acidic~~ Oxidising agent

(b) bad

(c) acidic

(d) Lead chloride

(e) F

(iii) $SO_2 = 32 + 32 = 64g$ $64g \text{ of } SO_2 \longrightarrow 22.4$ $1g \text{ of } SO_2 \longrightarrow 224$ $640 \quad 112$ $320g \text{ of } SO_2 \longrightarrow 224 \times 320$ 640 2

$$= 112 \text{ l}$$

(ii) equal volumes of all gases under same conditions of temp. and pressure contains equal no. of molecules.

(iii) Mass of propane = 8.8
 No. of moles = $\frac{\text{mass}}{\text{molar mass}} = \frac{8.8}{44} = 0.2$
 $0.2 \text{ moles} \rightarrow 22.4$
 $1 \text{ mole} \rightarrow 22.4$

16g of $\text{O}_2 \rightarrow 22.4$
 1g of $\text{O}_2 \rightarrow \frac{22.4}{16}$
 $2.2 \text{ g} \rightarrow \frac{22.4}{16} \times 2.2 = 3.08 \text{ l}$

(iv) ii) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
 $64 + 32 + 64 + 5 \times 18$
 $160 + 90$
 $= 250 \text{ g}$
 $= \frac{90}{250} \times 100 = 36\%$

(iii) X_2Y_2 n
 Molecular formula = $\frac{2 \times \text{X}_2\text{Y}_2}{\text{X}_2\text{Y}_2} = 2$
 $\Rightarrow \text{X}_2\text{Y}_4$

(i) Na = 23
 $23 \text{ g of Na} \rightarrow \text{Na}$
 $1 \text{ g of Na} \rightarrow \frac{\text{Na}}{23}$
 $4.6 \text{ g of Na} \rightarrow \frac{\text{Na}}{23} \times 4.6 = 0.2 \times \text{Na}$

(v) 2 = He 4 = Be 8 = O 15 = P 19 = K
 (i) 15 = P
 19 = K

(iii) 8 = O
 (iv) 4 = Be
 (v) 2 = He

SEC

2. (i) 1

(ii)

4

2

(iii)
in
i

(iii) $8 = 0$

(iv) $4 = \text{Be}$

(v) $2 = \text{He}$

SECTION B

Q 2. (i) 1) T
2) T

(ii) CO_2

$$12 + 32 = 44 \text{ g}$$

44 g of $\text{CO}_2 \longrightarrow \text{NA}$

1 g of $\text{CO}_2 \longrightarrow \frac{\text{NA}}{44}$

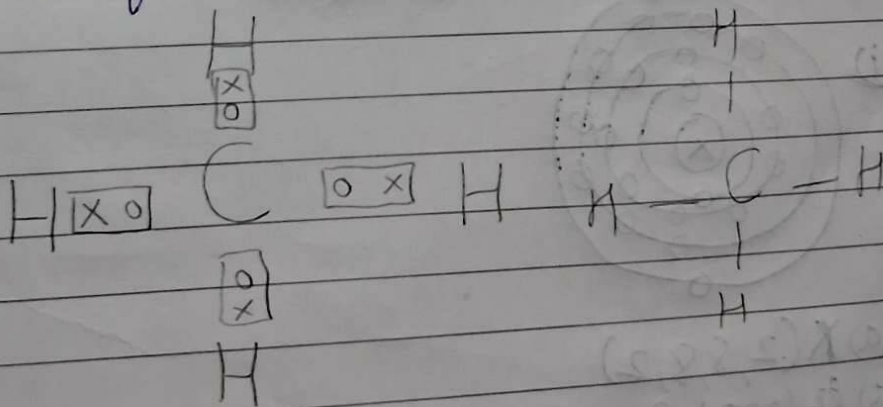
22 g of $\text{CO}_2 \longrightarrow \frac{\text{NA}}{44} \times 22 = 0.5 \times \text{NA}$

(iii) (1) Less because as we go left to right the electron affinity increases and ~~also~~ electron charge also increases due to which it ~~is~~ is less

2) ~~More~~ More

3) right side of Y

(iv) (i)



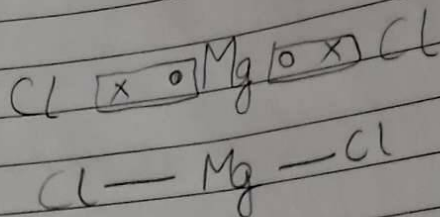
(ii) (a)

(b) All elements valence electrons contain 2 electrons

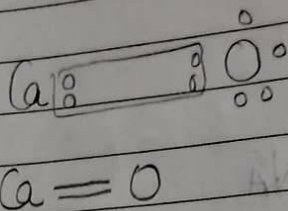
Q 3. (i)

(ii) This is because that both have different electronegativity.
(b) because there are 4 chlorine present in the bond.

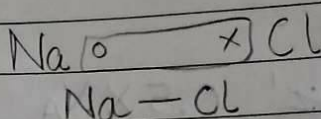
(i)



(ii)



(iii)



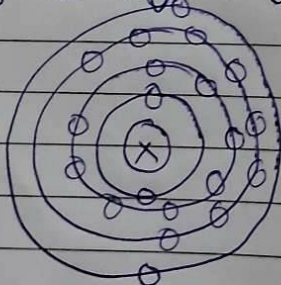
(iv) bond formed by a shared pair of electron with both electrons coming from the same atom.

It has properties of both ionic & covalent bond

There should be lone pairs to form the bond.

eg: ~~H_2O~~ to ~~NH_3~~ to form NH_4

Q4. (i)



(a) ~~X~~ (2, 8, 8, 2)

(b) ~~Ca~~ Group 2 and Period 3

(ii) (a) 2

(b) ionic compound

(iii)

- (iv) (i) A and B C and D belong to same period
(ii) A and C B and D belong to same group
(iii) B and D belong to 18th group

6. (i) (a) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
(b) NaHS

(ii) -

(iii) They are
> Three Ph
> acid + base

(iv) (a) 3

(b) 2

(c) 1

(d) 1

(e) 2

(f) 1

Q 6. (i) (a) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
(b) NaHSO_4

(ii) —

(iii) > They are sour in taste

> Their pH is less than 7

> acid + base gives salt and water

(iv) (a) 3

(b) 2

(c) 1

(d) 1

(e) 2

(f) 1